



Tennessee Department of Transportation  
Aeronautics Division  
7335 Centennial Boulevard  
Nashville, Tennessee 37209  
615-741-3208

<https://www.tn.gov/tdot/aeronautics.html>

## Overview:

The Tennessee Department of Transportation Aeronautics Division (Aeronautics) developed a framework that can be used by Aeronautics and its consultants to complete pavement rehabilitation alternatives. LCCA can be a valuable tool for comparing alternatives. The use of a LCCA methodology helps to shift the focus in project selection and toward those which have the lowest costs over the life of the pavement. The LCCA cycle cost estimates for TN airport pavement reconstruction projects to assist in the selection of the most cost-effective alternative. LCCA is not considered a complete Engineer's Estimate of Probable Costs.

Applied Pavement Technology, Inc. (APTech) has worked with Aeronautics to reconstruction alternatives to assist with evaluating the most cost-effective the guidelines provided in the Airfield Asphalt Pavement Technology Program study (AAPTTP Project 06-06) and Federal Aviation Administration (FAA) guidance of this release), *Airport Pavement Design and Evaluation*. The developed framework for future maintenance activities and their costs, intermediate rehabilitation activities, and of daily operation revenue associated with pavement closures to generate

Maintenance policies and timing have been based on analysis of data collected from the Management Program (PMP) and annual maintenance program. As pavement conditions deteriorate through the PMP, policies will be revisited. Pavement closure associated loss and cost alternatives. The analysis period has been set to 30 years based on review of

Depending on the durations maintenance activities, the framework has been limited facility uses or closures to implement repairs and rehabilitation.

Salvage value for the pavement alternatives is based on the assumed remaining value, the monetary calculation of the worth of the existing pavement. Adding value to recycling pavement materials at the end of their life, there is a suitability of the material for reuse, the future demand for the recycled materials that residual value is not used in these calculations.

Additional analysis inputs include discount rate and contingency fees (mobil established by Aeronautics and should only be revised after discussion with

If interested in the full report produced as part of developing this LCCA fram

tics) has sponsored the development of an analytical a reasonable life-cycle cost analysis (LCCA) of comparing alternatives of varying costs and service lives. away from alternatives with the lowest initial costs . This workbook assists in the development of life-cost with pavement type selection. It is not to be

develop this LCCA framework to compare feasible options. The developed LCCA framework considers n's *Life Cycle Cost Analysis for Airport Pavements* guidance in Advisory Circular 150/5320-6G (at the time framework considers construction (or initial) costs, activities and associated costs, salvage value and loss a Present Value for the alternatives.

ted as part of the Tennessee Statewide Pavement performance data continues to be collected ; of revenue is accounted for as indirect cost of d TN PMP and nationwide data.

n updated to account for lost revenue associated with

ining life as a percentage of construction cost. ent at the end of the service life, while certainly are enough uncertainties associated with the erial, and the costs associated with recycling the

ization and engineering). These have been Aeronautics.

ework, contact Aeronautics.



**TDOT**

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Transportation

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#### Instructions:

This workbook has been set up to automate maintenance cost calculations and analyses for Aeronautics' review. This workbook contains privileged information by Aeronautics.

The use of the workbook is fairly straightforward, and the user should be able to generate a life-cycle cost estimate. Excluding this "Instructions" worksheet and "Overview" worksheets: "General Information," "Pay Items," "Maintenance Policies," and "Summary" worksheets are for reference only and should only be altered after discussion with Aeronautics. Additional worksheets for anticipated reconstruction alternatives are also included.

The following step-by-step procedures are provided to assist in using the workbook. Be sure to follow these directions in the logical, step-by-step fashion. Do not use the "Format" menu feature as this can write over any functions, commands, or equations. Make sure that functions work properly. There have also been issues with functions in the past. Opening the file in the main monitor should correct function issue, if they occur. If you see the message "CONTENT: The ActiveX content in this file is blocked", go to File > Options > Security > select "Prompt me before enabling all controls with minimal restrictions", click "Yes", and the content will be greyed out, ask your IT department to allow ActiveX for this file.

Note: Throughout the workbook, cells that require user-supplied inputs are indicated by a yellow background (and should not be modified by users).

1. Go to the "General Information" worksheet and enter in the relevant information (such as project airport, branch information, pavement area, and so on) that will be used for computations. Ensure the appropriate selection is made in cell D38.

2. Still in the "General Information" worksheet, click on the "Alternative Setup" button. A form will pop up for adding alternatives. The pop-up window includes the list of alternatives that will be used, update details and description, if desired, and click "Add" to add new alternatives. Alternatives can also be reorganized using the buttons "Move Up" and "Move Down" and a worksheet for each alternative will be created. A "Summary" worksheet will also be created.

3. Modify existing Pay Items, or insert new items in Table2 in the Pay\_Items worksheet. The Pay\_Items worksheet includes paving items only and does not include other items, such as electrical, drainage, etc.

4. To include lost airport revenue during runway closures, set cell D38 on the "General Information" worksheet to "Include Lost Revenue". The average daily revenue for the selected airport is pulled automatically from the Airport Revenue worksheet which covers the 17 airports meeting Aeronautics' activity criteria (cell D39 on the "General Information" worksheet reads "Missing Airport Revenue", lost revenue is carried as \$0, and a warning message is displayed to Aeronautics to add revenue data. To view the hidden worksheet: Home > Formulas > Show Hidden Worksheets.

5. Go to the worksheet for each alternative included in the LCCA and fill in the quantities for each pay item (including accounting for indirect costs). Select the appropriate pay items in cells C13 through F10 (if a pay item is not needed; leave extra cells blank). Enter the quantities for each selected pay item. The quantities for pay items C13 through F10 are pre-filled from production-rate defaults (surface treatment 1 and slab work) and may be overridden with project-specific values.

6. Once the pay items and quantities have been filled in, the Net Present Worth (NPW) will be calculated for each alternative. The NPW is displayed in cell G11 of the worksheet for each alternative.

should be automatically filled in for the alternative. The "Summary" worksheet is updated once all individual alternative worksheets are completed. The Summary worksheet summarizes the costs for construction, maintenance, rehabilitation, lost revenue and salvage present in the alternative, the lowest-cost alternative and whether it still wins at the FAA AIP rate of 7%, and the impact of any changes. Use the "View Summary" button under Alternative Setup to open it.

If there are any issues or questions with the use of the LCCA workbook, please contact the LCCA team.

so as to expedite the overall preparation of life-cycle cost  
tion and is intended for use only by consultants authorized

le to easily work their way through the development of a  
ew" worksheet, the workbook initially contains four  
l "Summary," The Pay Items and Maintenance Policies  
n with Aeronautics. From the General Information  
are created.

rkbook and in understanding how the series of worksheets  
shion to avoid problems. Also, don't use the cut and paste  
sure to "Enable Content" in Excel when the file is opened  
running properly when multiple monitors are in use.  
cur. If Excel shows a yellow bar reading "BLOCKED  
Trust Center > Trust Center Settings > ActiveX Settings,  
ck OK, then close and reopen Excel. If that setting is

ndicated in grey shading. Most other cells are calculated or

s in cells D9 through D39. This is general information  
is needed to identify the project and complete the cost

ip" button at the bottom of the information list. A separate  
t of available alternatives for this LCCA. Select which  
k on "Add." The form can also be used to "Copy" or  
s at the bottom of the form. Once completed, click "Close"  
et is also populated as alternatives are completed.

worksheet as needed. Note that this LCCA is intended for  
iage, and so on.

e "General Information" worksheet to Yes before creating  
omatically from the hidden RevenueData worksheet,  
shows the value). If the airport is not in that list, D39  
g appears on each alternative worksheet; contact  
rmat > Hide & Unhide > Unhide Sheet.

ells C4 through C8, and F4 through F10 (for analyses  
through C22 for the alternative (only select the cells you  
m in cells E13 through E22. Closure durations in cells F4  
15,000 SY/day, mill and overlay 3,800 SY/day, PCC joint

orth (NPW) table and summary graph below the pay items

set will also automatically fill with alternative costs as the  
shows a results table for each alternative (initial  
worth, net present worth, closure days), a line naming the  
and five charts. It updates by itself whenever an input  
t and the buttons at the top of the Summary to return.

se contact Aeronautics.



#### HOW TO USE THIS WORKBOOK

1. Fill in the grey cells below (D9 to D39).
  2. Click Alternative Setup and add your alternatives.
  3. Click View Summary to compare them.
- Grey cells are yours to fill in; white cells calculate themselves.  
Typical Values lists the usual range for each input.

#### Airport Information:

|                    |     |
|--------------------|-----|
| Airport Name:      |     |
| Airport ID:        | N/A |
| City:              | N/A |
| Airport Owner:     | N/A |
| State Region:      | N/A |
| Design Consultant: |     |
| Project Number:    |     |
| Project Name:      |     |
| Date of LCCA:      |     |

Na

**Still needed: Airport Name, Construction Year, Mainline Area, Markings Area**

#### Project Information:

|                            |  |
|----------------------------|--|
| Branch Type:               |  |
| Branch Name:               |  |
| Project Type:              |  |
| Project Description:       |  |
| Initial Construction Year: |  |
| Mainline Area, S.Y.:       |  |
| Shoulder Area, S.Y.:       |  |
| Markings, S.F.:            |  |
| Markings, Type:            |  |

Construction year; the analysis period starts here.  
Length x width / 9. TN GA: 4,000-6,000 ft x 75-100 ft wide.  
Paved shoulders only; usually 0 at TN GA airports.  
About 2,000-3,000 SF per 1,000 ft of runway.

#### LCCA Parameters:

|                                |                                                                       |
|--------------------------------|-----------------------------------------------------------------------|
| Analysis Period, years:        | 30                                                                    |
| Discount Rate, percent:        | 3                                                                     |
| Salvage Value, \$:             | Remaining service life at end of analysis period (as percent of cost) |
| Mobilization, percent:         | 10                                                                    |
| Engineering, percent:          | 5                                                                     |
| Account Indirect Cost (Yes/No) | No                                                                    |
| Airport Daily Revenue          |                                                                       |

TDOT policy 30 years; FAA leaves the period to the engineer.  
TDOT 3%. FAA uses the OMB A-94 real rate, 2.0% in 2026.  
Percent of pay items; workbook default 10.  
Percent of pay items; now applied to initial construction too.  
Yes counts lost revenue during closures (17 airports have data).

#### Alternative Setup:

Alternative Setup

#### LCCA Summary:

View Summary ►

#### Typical input values:

Typical Values ►

| Division                                | Division Type          | Pay Item No. |
|-----------------------------------------|------------------------|--------------|
| <b>Part 3 – Sitework</b>                | Default                |              |
|                                         | Sitework               | P-101-5.1a   |
|                                         | Sitework               | P-101-5.1b   |
|                                         | Sitework               | P-101-5.2a   |
|                                         | Sitework               | P-101-5.2b   |
|                                         | Sitework               | P-101-5.2c   |
|                                         | Sitework               | P-101-5.2d   |
|                                         | Sitework               | P-101-5.3    |
|                                         | Sitework               | P-101-5.4a   |
|                                         | Sitework               | P-101-5.4b   |
|                                         | Sitework               | P-101-5.5a   |
|                                         | Sitework               | P-101-5.5b   |
|                                         | Sitework               | P-101-5.5c   |
|                                         | Sitework               | P-101-5.6    |
|                                         | Sitework               | P-152-4.1    |
|                                         | Sitework               | P-152-4.2    |
|                                         | Sitework               | P-154-5.1    |
|                                         | Sitework               | P-154-5.2    |
|                                         | Sitework               | P-155-8.1    |
|                                         | Sitework               | P-156-8.1    |
|                                         | Sitework               | P-157-8.1a   |
|                                         | Sitework               | P-157-8.1b   |
|                                         | Sitework               | P-158-8.1    |
| <b>Part 4 – Base Courses</b>            | Base Course            | P-207-5.1    |
|                                         | Base Course            | P-208-5.1    |
|                                         | Base Course            | P-208-5.2    |
|                                         | Base Course            | P-209-5.1    |
|                                         | Base Course            | P-209-5.2    |
|                                         | Base Course            | P-219-5.1    |
|                                         | Base Course            | P-219-5.2    |
|                                         | Base Course            | P-220-6.1    |
| <b>Part 5 – Stabilized Base Courses</b> | Stabilized Base Course | P-304-8.1    |
|                                         | Stabilized Base Course | P-306-8.1    |
|                                         | Stabilized Base Course | P-307-7.1    |
| <b>Part 6 – Flexible Pavements</b>      | Flexible Pavement      | P-401-8.1a   |
|                                         | Flexible Pavement      | P-401-8.1b   |
|                                         | Flexible Pavement      | P-401-8.1c   |
|                                         | Flexible Pavement      | P-403-8.1a   |
|                                         | Flexible Pavement      | P-403-8.1b   |
|                                         | Flexible Pavement      | P-403-8.1c   |
|                                         | Flexible Pavement      | P-407-X.1    |
| <b>Part 7 – Rigid Pavement</b>          | Cement Concrete        | P-501-8.1a   |
|                                         | Cement Concrete        | P-501-8.1b   |
|                                         | Cement Concrete        | P-501-8.1c   |
|                                         | Cement Concrete        | P-501-8.1d   |
|                                         | Surface Treatment      | P-608-8.1    |
| <b>Part 8 – Surface Treatments</b>      | Surface Treatment      | P-623-8.1    |
|                                         | Surface Treatment      | P-626-6.1    |
|                                         | Surface Treatment      | P-629-8.1    |
|                                         | Surface Treatment      | P-630-7.1a   |
|                                         | Surface Treatment      | P-631-7.1b   |
|                                         | Surface Treatment      | P-631-7.1b   |
| <b>Part 9 – Miscellaneous</b>           | Tack Coat              | P-603-5.1    |
|                                         | Joint Sealant          | P-605-5.1a   |
|                                         | Marking                | P-620-5.1a   |
|                                         | Marking                | P-620-5.1b   |



Grooving  
Grooving

P-621-5.1A  
P-621-5.1B

## **Pay Item Description**

Pavement Removal, Asphalt  
Pavement Removal, Concrete  
Crack Repair (Type I), Asphalt  
Crack Repair (Type II), Asphalt  
Crack Seal, Concrete  
Crack Repair, Concrete  
Markings Removal  
Spall Repair, Asphalt  
Full-Depth Repair, Asphalt  
Spall Repair, Concrete  
Full-Depth Repair, Concrete  
Slab Replacement, Concrete  
Cold Milling  
Unclassified Excavation  
Embankment In-Place  
Subbase Course  
Separation Geotextile (P-154)  
Lime Treated subgrade  
Cement-Treated Subgrade  
Cement Kiln Dust-Treated Subgrade  
Lime Kiln Dust-Treated Subgrade  
Fly Ash-Treated Subgrade  
In-Place Full Depth Reclamation (FDR) Recycled Asphalt Aggregate Base Course  
Aggregate Base Course  
Separation Geotextile (P-208)  
Crushed Aggregate Base Course  
Separation Geotextile (P-209)  
Recycled Concrete Aggregate Base Course  
Separation Geotextile (P-219)  
Cement-Treated Soil Base Course  
Cement-Treated Base Course  
Lean Concrete Base Course  
Cement-Treated Permeable Base Course (CTPB)  
Asphalt Surface Course  
Asphalt Base Course  
Asphalt Leveling Course  
Asphalt Mixture Surface Course  
Asphalt Mixture Base Course  
Asphalt Mixture Leveling Course  
Asphalt-Treated Permeable Base Course (ATPB)  
Concrete Pavement, 5-inch  
Concrete Pavement, 6-inch  
Concrete Pavement, 9-inch  
Concrete Pavement, 11-inch  
Emulsified Asphalt Surface Treatment  
Emulsified Asphalt Spray Seal Coat  
Emulsified Asphalt Slurry Seal Surface Treatment  
Thermoplastic Coal Tar Emulsion Surface Treatment  
Refined Coal Tar Emulsion Without Additives, Slurry Seal Surface Treatment  
Refined Coal Tar Emulsion With Additives, Slurry Seal Surface Treatment  
Emulsified Asphalt Tack Coat  
Joint Clean and Reseal, Concrete  
Markings (prep, markings, reflective media)  
Markings (prep, markings, non-reflective media)

Grooving, Asphalt  
Grooving, Concrete

| Units | Unit Cost | Average Pay Item Unit Cost |      |                                |
|-------|-----------|----------------------------|------|--------------------------------|
|       |           | Middle                     | West | East                           |
| N/A   | N/A       |                            |      |                                |
| S.Y.  | \$4.20    |                            |      | Thickness adjustment or mill   |
| S.Y.  |           |                            |      | Thickness adjustment or mill   |
| L.F.  | \$4.00    |                            |      |                                |
| L.F.  | \$6.50    |                            |      |                                |
| L.F.  | \$8.00    |                            |      |                                |
| L.F.  | \$8.50    |                            |      |                                |
| S.F.  | \$0.78    |                            |      |                                |
| S.F.  | \$8.50    |                            |      | Thickness adjustment or mill   |
| S.F.  |           |                            |      | Assumed Section: 4-inch H      |
| S.F.  | \$60.00   |                            |      | Thickness adjustment or mill   |
| S.Y.  | \$120.00  |                            |      | Thickness adjustment or mill   |
| S.Y.  | \$100.00  |                            |      |                                |
| S.Y.  | \$5.00    |                            |      | Thickness adjustment or mill   |
| C.Y.  |           |                            |      |                                |
| C.Y.  |           |                            |      |                                |
| C.Y.  | \$55.00   |                            |      |                                |
| S.Y.  |           |                            |      |                                |
| S.Y.  | \$10.00   |                            |      | Thickness adjustment or mill   |
| S.Y.  |           |                            |      | Thickness adjustment or mill   |
| S.Y.  |           |                            |      | Thickness adjustment or mill   |
| S.Y.  |           |                            |      | Thickness adjustment or mill   |
| S.Y.  |           |                            |      | Thickness adjustment or mill   |
| S.Y.  |           |                            |      | Thickness adjustment or mill   |
| C.Y.  |           |                            |      |                                |
| S.Y.  |           |                            |      |                                |
| C.Y.  | \$65.00   |                            |      |                                |
| S.Y.  |           |                            |      |                                |
| C.Y.  |           |                            |      |                                |
| S.Y.  |           |                            |      |                                |
| S.Y.  |           |                            |      | Thickness adjustment or mill   |
| C.Y.  |           |                            |      | If S.Y., then thickness adjust |
| C.Y.  |           |                            |      | If S.Y., then thickness adjust |
| C.Y.  |           |                            |      | If S.Y., then thickness adjust |
| TON   | \$130.00  |                            |      |                                |
| TON   | \$120.00  |                            |      |                                |
| TON   | \$110.00  |                            |      |                                |
| TON   |           |                            |      |                                |
| TON   |           |                            |      |                                |
| TON   |           |                            |      |                                |
| TON   |           |                            |      |                                |
| S.Y.  | \$79.00   |                            |      | Thickness adjustment or mill   |
| S.Y.  | \$82.00   |                            |      | Thickness adjustment or mill   |
| S.Y.  | \$85.00   |                            |      | Thickness adjustment or mill   |
| S.Y.  | \$91.00   |                            |      | Thickness adjustment or mill   |
| S.Y.  | \$4.00    |                            |      |                                |
| S.Y.  | \$5.00    |                            |      |                                |
| S.Y.  | \$7.00    |                            |      |                                |
| S.Y.  |           |                            |      |                                |
| S.Y.  |           |                            |      |                                |
| S.Y.  |           |                            |      |                                |
| GAL   | \$4.50    |                            |      | Assume 0.1 gal/SY for mill ;   |
| L.F.  | \$7.00    |                            |      | Assume 12.5x12.5 slabs for     |
| S.F.  | \$0.67    |                            |      |                                |
| S.F.  |           |                            |      |                                |

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**Table 1. New HMA maintenance policies.**

|                            |
|----------------------------|
| <b>Asphalt Maintenance</b> |
| <b>Maintenance 1</b>       |
| <b>Maintenance 2</b>       |
| <b>Maintenance 3</b>       |
| <b>Maintenance 4</b>       |
| <b>Rehabilitation 1</b>    |
| <b>Maintenance 5</b>       |
| <b>Maintenance 6</b>       |
| <b>Salvage</b>             |

**Table 2. New PCC maintenance policies.**

|                             |
|-----------------------------|
| <b>Concrete Maintenance</b> |
| <b>Maintenance 1</b>        |
| <b>Rehabilitation 1</b>     |
| <b>Salvage</b>              |

**Table 3. First HMA Overlay maintenance po**

|                            |
|----------------------------|
| <b>Asphalt Maintenance</b> |
| <b>Maintenance 1</b>       |
| <b>Maintenance 2</b>       |

|                         |
|-------------------------|
| <b>Maintenance 3</b>    |
| <b>Rehabilitation 2</b> |
| <b>Maintenance 4</b>    |
| <b>Maintenance 5</b>    |
| <b>Maintenance 6</b>    |
| <b>Salvage</b>          |

**Table 4. PCC rehabilitation maintenance po**

|                             |
|-----------------------------|
| <b>Concrete Maintenance</b> |
| <b>Maintenance 1</b>        |
| <b>Rehabilitation 1</b>     |
| <b>Salvage</b>              |



| Maintenance Item                                                   | Rate   |
|--------------------------------------------------------------------|--------|
| Surface treatment, SYD                                             | 1.0000 |
| Paint markings, SFT                                                | 1.0000 |
| Surface treatment, SYD                                             | 1.0000 |
| Paint markings, SFT                                                | 1.0000 |
| Patching: 0.75% of area                                            | 0.0075 |
| Crack sealing: 3.25% of area as linear feet                        | 0.0325 |
| Surface treatment: 100% of area                                    | 1.0000 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Patching: 1.00% of area                                            | 0.0100 |
| Crack sealing: 3.50% of area as linear feet                        | 0.0350 |
| Surface treatment: 100% of area                                    | 1.0000 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Mill and overlay: 4-inch cold mill, tack coat, and asphalt overlay | 1.0000 |
| Pre-overlay crack repair: 1.5% of area as linear feet              | 0.0150 |
| Pre-overlay patching: 0.5% of area                                 | 0.0050 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Surface treatment: 100% of area                                    | 1.0000 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Patching: 1.25% of area                                            | 0.0125 |
| Crack sealing: 4.0% of area as linear feet                         | 0.0400 |
| Surface treatment: 100% of area                                    | 1.0000 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Salvage value: 2 years (0.125) of mill and overlay                 | 0.1250 |

| Maintenance Item                                       | Rate   |
|--------------------------------------------------------|--------|
| Joint resealing: 100% of joints                        | 1.0000 |
| Crack sealing: 0.25% of area as linear feet            | 0.0025 |
| Partial-depth patching: 0.13% of area                  | 0.0013 |
| Full-depth patching: 0.30% of area                     | 0.0030 |
| Slab replacements: 2% of area                          | 0.0200 |
| Full-depth patching: 0.30% of area                     | 0.0030 |
| Partial-depth patching: 0.13% of area                  | 0.0013 |
| Crack sealing: 0.30% of area as linear feet            | 0.0030 |
| Joint sealing: 100% of joints                          | 1.0000 |
| Salvage value: 10 years (0.25) of initial construction | 0.2500 |

**licies.**

| Maintenance Item                           | Rate   |
|--------------------------------------------|--------|
| Surface treatment: 100% of area            | 1.0000 |
| Paint markings: 100% of markings           | 1.0000 |
| Patching: 1.25% of area                    | 0.0125 |
| Crack sealing: 4.0% of area as linear feet | 0.0400 |
| Surface treatment: 100% of area            | 1.0000 |
| Paint markings: 100% of markings           | 1.0000 |

|                                                                    |        |
|--------------------------------------------------------------------|--------|
| Surface treatment, SYD                                             | 1.0000 |
| Paint markings, SFT                                                | 1.0000 |
| Mill and overlay: 4-inch cold mill, tack coat, and asphalt overlay | 1.0000 |
| Pre-overlay crack repair: 2% of area as linear feet                | 0.0150 |
| Pre-overlay patching: 1% of area                                   | 0.0050 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Surface treatment: 100% of area                                    | 1.0000 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Patching: 1.25% of area                                            | 0.0125 |
| Crack sealing: 4.0% of area as linear feet                         | 0.0400 |
| Surface treatment: 100% of area                                    | 1.0000 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Surface treatment: 100% of area                                    | 1.0000 |
| Paint markings: 100% of markings                                   | 1.0000 |
| Salvage value: Need for reconstruction (zero salvage)              | 0.0000 |

**licies.**

| <b>Maintenance Item</b>                               | <b>Rate</b> |
|-------------------------------------------------------|-------------|
| Joint resealing: 100% of joints                       | 1.0000      |
| Crack sealing: 0.35% of area as linear feet           | 0.0035      |
| Partial-depth patching: 0.2% of area                  | 0.0020      |
| Full-depth patching: 0.30% of area                    | 0.0030      |
| Slab replacements: 5% of area                         | 0.0500      |
| Full-depth patching: 0.30% of area                    | 0.0030      |
| Partial-depth patching: 0.15% of area                 | 0.0015      |
| Crack sealing: 0.30% of area as linear feet           | 0.0030      |
| Joint sealing: 100% of joints                         | 1.0000      |
| Salvage value: Need for reconstruction (zero salvage) | 0.0000      |

| Year<br>Applied |
|-----------------|
| 4               |
| 8               |
| 12              |
| 16              |
| 20              |
| 24              |
| 28              |
| 30              |

| Year<br>Applied |
|-----------------|
| 19              |
| 27              |
| 30              |

| Year<br>Applied |
|-----------------|
| 4               |
| 8               |

|    |
|----|
| 12 |
| 16 |
| 20 |
| 24 |
| 28 |
| 30 |

| Year<br>Applied |
|-----------------|
| 15              |
| 25              |
| 30              |

| Alternative | Name | Initial Construction | Present Worth | Alternative Description |
|-------------|------|----------------------|---------------|-------------------------|
|-------------|------|----------------------|---------------|-------------------------|

LCCA SUMMARY

Everything from this column onward calculates from the alternative worksheets and updates by itself. Columns A-E and "Chart 1" are written by the Alternative Setup form as below.

| Worksheet | Alternative | Type | Initial construction | Maintenance PW | Rehabilitation PW | Lost revenue PW | Salvage PW | Net present worth | vs. lowest NPW | Closure days in period | Runway availability |
|-----------|-------------|------|----------------------|----------------|-------------------|-----------------|------------|-------------------|----------------|------------------------|---------------------|
|           |             |      |                      |                |                   |                 |            |                   |                |                        |                     |
|           |             |      |                      |                |                   |                 |            |                   |                |                        |                     |
|           |             |      |                      |                |                   |                 |            |                   |                |                        |                     |

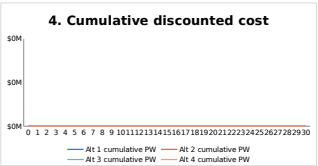
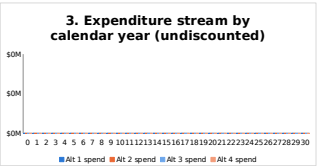
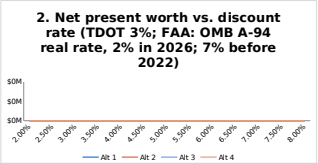
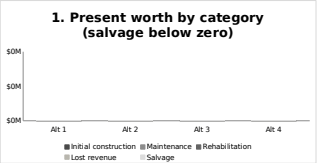
No alternatives yet. Go to General Information and click Alternative Setup.

COMPARISON (RealCost layout: agency cost, user cost, total; present worth and equivalent uniform annual cost)

| Alternative | Agency cost PW | Agency EUAC | User cost PW (lost revenue) | User EUAC | Total PW | Total EUAC | vs. lowest (\$) | vs. lowest (%) | Lowest? |
|-------------|----------------|-------------|-----------------------------|-----------|----------|------------|-----------------|----------------|---------|
|             |                |             |                             |           |          |            |                 |                |         |
|             |                |             |                             |           |          |            |                 |                |         |
|             |                |             |                             |           |          |            |                 |                |         |

Agency cost = initial construction + maintenance + rehabilitation + salvage. User cost = lost airport revenue during runway closures, the airport-side counterpart of the user delay cost in the CaltransFTRisk RealCost layout. EUAC =  $PW \times (1+i)^N / [(1+i)^N - 1]$  at the discount rate and analysis period on General Information. "vs. lowest (%)" is the difference as a share of the lowest total PW.

CHARTS



How to read: 1 shows where each alternative's cost sits; 2 whether the lowest-cost alternative holds at other discount rates (FAA now uses the OMB A-94 real rate, about 2%; 7% was the rule before 2022); 3 and 4 when the money is spent and when the higher first cost is paid back; 5 how often and how long the runway closes.

Reference sheet: nothing here feeds the calculation.

**TYPICAL VALUES AND HELPERS FOR THE INPUTS**

Reference only: nothing on this sheet feeds the calculation. Use it to sanity-check what you type on General Information and the alternative worksheets. Values are ranges from the cited sources as of the date in the source column; confirm against current bid tabulations for a specific project.

| 1. What TDOT Aeronautics manages (context for project size)                                   |                                                      |                |      |  |  |                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------|------------------------------------------------------|----------------|------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Item                                                                                          | Value                                                | Unit           | Year |  |  | Source                                                                                                                                                      |
| Public-use airports in the Tennessee system (TASP 2040)                                       |                                                      | 78 airports    | 2021 |  |  | <a href="https://www.tasp2040.com/wp-content/uploads/2021/03/Updated-TASP-Inven">https://www.tasp2040.com/wp-content/uploads/2021/03/Updated-TASP-Inven</a> |
| Commercial service / general aviation                                                         | 6 / 72                                               | airports       | 2021 |  |  | <a href="https://www.tasp2040.com/wp-content/uploads/2021/03/Updated-TASP-Inven">https://www.tasp2040.com/wp-content/uploads/2021/03/Updated-TASP-Inven</a> |
| Airports in the FAA NPIAS (eligible for AIP)                                                  |                                                      | 69 airports    | 2021 |  |  | <a href="https://www.tasp2040.com/wp-content/uploads/2020/12/Chapter-7-Airport-">https://www.tasp2040.com/wp-content/uploads/2020/12/Chapter-7-Airport-</a> |
| Airports in the TDOT pavement management network (APTech APMS, PCI per ASTM D5340)            | ~70                                                  | airports       | 2022 |  |  | <a href="https://www.appliedpavement.com/tennessee-apms">https://www.appliedpavement.com/tennessee-apms</a>                                                 |
| PCI performance objective, runways / taxiways                                                 | 78 / 75                                              | PCI            | 2021 |  |  | <a href="https://www.tn.gov/content/dam/tn/dot/aeronautics/TN%20Airport%20Mana">https://www.tn.gov/content/dam/tn/dot/aeronautics/TN%20Airport%20Mana</a>   |
| TASP system-wide estimate: pavement rehabilitation (10-yr life goal)                          |                                                      | 70,865,874 \$  | 2021 |  |  | <a href="https://www.tasp2040.com/wp-content/uploads/2021/09/TASP-Cost-Estimate">https://www.tasp2040.com/wp-content/uploads/2021/09/TASP-Cost-Estimate</a> |
| TASP system-wide estimate: pavement new / reconstruction (20-yr life goal)                    |                                                      | 173,117,648 \$ | 2021 |  |  | <a href="https://www.tasp2040.com/wp-content/uploads/2021/09/TASP-Cost-Estimate">https://www.tasp2040.com/wp-content/uploads/2021/09/TASP-Cost-Estimate</a> |
| State share on GA capital projects (federal or state funded)                                  | up to 90% (10% local); 95/5 on federal discretionary |                | 2023 |  |  | <a href="https://www.tn.gov/content/dam/tn/dot/documents/170-02.pdf">https://www.tn.gov/content/dam/tn/dot/documents/170-02.pdf</a>                         |
| Statewide Airfield Pavement and Marking Maintenance Program (crack seal, seal coat, markings) | 100% state funded, no local share                    |                | 2024 |  |  | <a href="https://www.tn.gov/dot/aeronautics/apmm-contract.html">https://www.tn.gov/dot/aeronautics/apmm-contract.html</a>                                   |
| Pavement areas and PCI by airport (statewide tables)                                          | see APTech IDEA portal                               |                | 2022 |  |  | <a href="https://idea.appliedpavement.com/hosting/tennessee/statewide-summary/i">https://idea.appliedpavement.com/hosting/tennessee/statewide-summary/i</a> |
| NPIAS 2025-2029 classification of each TN airport (National / Regional / Local / Basic)       | see Appendix A                                       |                | 2024 |  |  | <a href="https://www.faa.gov/sites/faa.gov/files/airports/planning_capacity/npi">https://www.faa.gov/sites/faa.gov/files/airports/planning_capacity/npi</a> |

The workbook dropdown holds the TASP airports (count in the live section below). The 17 airports with revenue data are those meeting Aeronautics' activity criteria.

| 2. Typical runway geometry at Tennessee GA airports (for D26 mainline area and D28 markings) |        |                            |                    |                    |                                                                                                                                                                                                 |                                                                                                                                                               |
|----------------------------------------------------------------------------------------------|--------|----------------------------|--------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Airport                                                                                      | Runway | Length x width (ft)        | Mainline area (SY) | Role               | Note                                                                                                                                                                                            | Source                                                                                                                                                        |
| Murfreesboro Municipal (MBT)                                                                 | 18/36  | 4,753 x 100                |                    | 52,811 GA, Middle  | workbook MBT run used 52,778 SY                                                                                                                                                                 | <a href="https://en.wikipedia.org/wiki/Murfreesboro_Municipal_Airport">https://en.wikipedia.org/wiki/Murfreesboro_Municipal_Airport</a>                       |
| Bomar Field, Shelbyville (SYI)                                                               | 18/36  | 5,503 x 100                |                    | 61,144 GA, Middle  |                                                                                                                                                                                                 | <a href="https://en.wikipedia.org/wiki/Shelbyville_Municipal_Airport_(Tennessee)">https://en.wikipedia.org/wiki/Shelbyville_Municipal_Airport_(Tennessee)</a> |
| Gatlinburg-Pigeon Forge (GKT)                                                                | 10/28  | 5,506 x 75                 |                    | 45,883 GA, East    | 75-ft width                                                                                                                                                                                     | <a href="https://en.wikipedia.org/wiki/Gatlinburg%E2%80%93Pigeon_Forge_Airport">https://en.wikipedia.org/wiki/Gatlinburg%E2%80%93Pigeon_Forge_Airport</a>     |
| Outlaw Field, Clarksville (CKV)                                                              | 17/35  | 5,999 x 100                |                    | 66,656 GA, Middle  | used in the new-project test                                                                                                                                                                    | <a href="https://en.wikipedia.org/wiki/Clarksville%E2%80%93Montgomery_County_Re">https://en.wikipedia.org/wiki/Clarksville%E2%80%93Montgomery_County_Re</a>   |
| McKellar-Sipes Regional (MKL)                                                                | 2/20   | 6,005 x 150                |                    | 100,083 GA, West   | rebuilt 2024, ~\$28M all-in                                                                                                                                                                     | <a href="https://www.wbbitv.com/2024/08/28/mckellar-sipes-regional-airport-set-">https://www.wbbitv.com/2024/08/28/mckellar-sipes-regional-airport-set-</a>   |
| Smyrna (MQY)                                                                                 | 1/19   | 5,546 x 100                |                    | 61,622 GA, Middle  | reconstructed 2025, ~\$13M all-in                                                                                                                                                               | <a href="https://www.smyrnaairport.com/news-events/smyrna-airport-runway-119-gr">https://www.smyrnaairport.com/news-events/smyrna-airport-runway-119-gr</a>   |
| Smyrna (MQY)                                                                                 | 14/32  | 8,038 x 150                |                    | 133,967 GA, Middle | former military                                                                                                                                                                                 | <a href="https://en.wikipedia.org/wiki/Smyrna_Airport_(Tennessee)">https://en.wikipedia.org/wiki/Smyrna_Airport_(Tennessee)</a>                               |
| Typical TN GA runway                                                                         |        | 4,000 to 6,000 x 75 or 100 | 33,000 to 67,000   |                    | TASP Regional Service minimum width 100 ft                                                                                                                                                      | <a href="https://www.tasp2040.com/wp-content/uploads/2020/12/Chapter-7-Airport-">https://www.tasp2040.com/wp-content/uploads/2020/12/Chapter-7-Airport-</a>   |
| Markings area rule of thumb                                                                  |        |                            |                    |                    | 2,000 to 3,000 SF per 1,000 ft of runway for a full runway marking set (centerline, thresholds, designators, aiming points, TDZ where applicable); use the marking plan quantity when available | engineering judgement; verify with the marking plan                                                                                                           |

Mainline area = length x width / 9. Shoulders (D27) are usually unpaved at TN GA airports; enter paved shoulders only. Area drives the closure-day defaults on the alternative sheets.

| 3. Unit costs: workbook defaults (Pay_Items, column F) against recent bids                             |                         |               |                                                    |       |                                                                                                        |                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------|-------------------------|---------------|----------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pay item                                                                                               | Workbook default (live) | Unit          | Recent bid range                                   | Year  | Reading                                                                                                | Source                                                                                                                                                      |
| P-401 Asphalt Surface Course                                                                           |                         | 130.00 \$/ton | 185 (low bid) to 232 (engineer's estimate)         |       | 2,025 Workbook default looks low against 2025 southeastern bids; confirm before relying on the default | <a href="https://aeronautics.sc.gov/sites/default/files/2025-05/250521_FLO_Reha">https://aeronautics.sc.gov/sites/default/files/2025-05/250521_FLO_Reha</a> |
| P-401 Asphalt Base Course                                                                              |                         | 120.00 \$/ton | typically 5 to 10% below surface course            |       |                                                                                                        | <a href="https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj">https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj</a> |
| P-501 Concrete Pavement, 9-inch                                                                        |                         | 85.00 \$/SY   | 165 (8-in doweled apron, planning value)           | 2,023 | Planning value from another state DOT; obtain a TN or southeastern PCC bid before use                  | <a href="https://www.dot.state.mn.us/aero/planning/documents/sasp/sasappendixf">https://www.dot.state.mn.us/aero/planning/documents/sasp/sasappendixf</a>   |
| P-209 Crushed Aggregate Base Course                                                                    |                         | 65.00 \$/CY   | not found in open sources; see SC bid tabs         |       |                                                                                                        | <a href="https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj">https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj</a> |
| P-154 Subbase Course                                                                                   |                         | 55.00 \$/CY   | not found in open sources; see SC bid tabs         |       |                                                                                                        | <a href="https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj">https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj</a> |
| P-101 Cold Milling                                                                                     |                         | 5.00 \$/SY    | 4.70 (3-in, low bid) to 10.00 (variable depth, EE) | 2,025 |                                                                                                        | <a href="https://aeronautics.sc.gov/sites/default/files/2025-05/250521_FLO_Reha">https://aeronautics.sc.gov/sites/default/files/2025-05/250521_FLO_Reha</a> |
| P-608 Emulsified Asphalt Surface Treatment                                                             |                         | 4.00 \$/SY    | TDOT APMM 2022 Summary of Bids                     | 2,022 | State maintenance contract prices apply to seal coats and crack sealing                                | <a href="https://www.tn.gov/dot/aeronautics/apmm-contract.html">https://www.tn.gov/dot/aeronautics/apmm-contract.html</a>                                   |
| P-620 Markings (reflective)                                                                            |                         | 0.67 \$/SF    | TDOT APMM 2022 Summary of Bids                     | 2,022 |                                                                                                        | <a href="https://www.tn.gov/dot/aeronautics/apmm-contract.html">https://www.tn.gov/dot/aeronautics/apmm-contract.html</a>                                   |
| All-in runway reconstruction, TN 2024-2025 (pavement + lighting + grading)                             |                         | \$/SY         | 210 (Smyrna 1/19) to 280 (MKL 2/20)                | 2,025 | Project-level figures; the LCCA compares pavement pay items only                                       | <a href="https://www.smyrnaairport.com/news-events/smyrna-airport-runway-119-gr">https://www.smyrnaairport.com/news-events/smyrna-airport-runway-119-gr</a> |
| State maintenance project example (Dyersburg): >200,000 LF crack seal, >121,000 SY seal coat, markings | 820,000 \$              |               |                                                    | 2,023 | about \$5/SY of runway treated                                                                         | <a href="https://www.stategazette.com/news/regional-airport-receives-state-gran">https://www.stategazette.com/news/regional-airport-receives-state-gran</a> |

Pay\_Items column F holds the default unit cost used by the alternative sheets (regional columns G-I are blank in v1.2.0). The defaults date from the 2022 framework; the 2025 comparison rows suggest the asphalt default is below current bids. Updating Pay\_Items is the only place a cost needs to change.

| 4. Closure-day defaults built into the alternative sheets (cells F4:F10; override with project-specific values) |                                |                                                                  |                       |                                            |  |                                  |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------|-----------------------|--------------------------------------------|--|----------------------------------|
| Operation                                                                                                       | Production rate in the formula | Days for a 6,000 x 100 ft runway (66,667 SY, 15,000 SF markings) | Where used            | Note                                       |  | Source                           |
| Surface treatment (seal coat)                                                                                   | 15,000 SY/day                  |                                                                  | 5 HMA Maintenance 1-6 | each operation is rounded up to whole days |  | this workbook (MBT and SRB runs) |

|                                                     |                                                                           |                                                                  |                                          |                                          |  |               |
|-----------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------|------------------------------------------|--|---------------|
| Markings: paint / layout and removal                | 20,000 SF/day and 50,000 SF/day                                           |                                                                  | 2 every HMA activity                     |                                          |  | this workbook |
| Patching                                            | 10,000 SY of pavement per day at the policy rate (0.75% to 1.25% of area) |                                                                  | 1 HMA Maintenance 3, 4, 6                |                                          |  | this workbook |
| Crack sealing                                       | 5,000 SY of pavement per day at the policy rate (3.25% to 4% of area)     |                                                                  | 1 HMA Maintenance 3, 4, 6                |                                          |  | this workbook |
| Mill and overlay (4 in)                             | 3,800 SY/day                                                              |                                                                  | 19 HMA Rehabilitation 1                  | the largest HMA closure                  |  | this workbook |
| PCC joint resealing                                 | about 1 LF of joint per SY of slab, 10,000 LF/day                         |                                                                  | 7 PCC Maintenance 1 and Rehabilitation 1 |                                          |  | this workbook |
| PCC crack sealing, partial- and full-depth patching | 5,000 to 10,000 SY of pavement per day at the policy rate                 |                                                                  | 3 PCC Maintenance 1                      |                                          |  | this workbook |
| PCC slab replacement (2% of area) + cure            | 1,000 SY/day plus 7 days cure                                             |                                                                  | 9 PCC Rehabilitation 1                   | the largest PCC closure                  |  | this workbook |
| Typical totals over 30 years                        |                                                                           | HMA 57 to 67 days in 7 closures; PCC 27 to 29 days in 2 closures | Summary column Q                         | MBT (52,778 SY) and CKV test (66,667 SY) |  | this workbook |

Initial construction is not a closure in the model (the runway is assumed closed for construction regardless of alternative). Lost revenue = closure days x airport daily revenue x 100%; a per-category "share lost" factor is a policy decision listed in the change notes.

##### 5. Economic parameters (D33 analysis period, D34 discount rate, salvage)

| Parameter                                                                     | Value                                                                              | Applies to                                 | Year |      | Reading                                                                                                                                                                 | Source                                                                                                                                                      |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------|------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TDOT policy discount rate (workbook default)                                  |                                                                                    | 3.00 %                                     |      |      | live from General Information D34                                                                                                                                       | this workbook                                                                                                                                               |
| TDOT policy analysis period (workbook default)                                |                                                                                    | 30.00 years                                |      |      | live from General Information D33                                                                                                                                       | this workbook                                                                                                                                               |
| FAA AIP: discount rate for pavement cost-effectiveness (PGL 22-01, June 2022) | OMB Circular A-94 Appendix C real rate for the analysis period                     | %                                          |      | 2022 | Replaced the fixed 7% rule; the analysis period is now the engineer's choice (the fixed 20-year requirement was removed)                                                | <a href="https://www.faa.gov/airports/aip/guidance_letters/aip_pgl_22_01">https://www.faa.gov/airports/aip/guidance_letters/aip_pgl_22_01</a>               |
| OMB A-94 Appendix C real rate, 20-year and 30-year (calendar 2026)            |                                                                                    | 2 %                                        |      | 2026 | Revised March 2026 (OMB memo M-26-09); 2025 values were 1.8% (10-yr) and 2.3% (30-yr)                                                                                   | <a href="https://www.whitehouse.gov/wp-content/uploads/2026/03/CircularA-94Appe">https://www.whitehouse.gov/wp-content/uploads/2026/03/CircularA-94Appe</a> |
| Legacy FAA AIP rule before PGL 22-01                                          | 7% and 20 years                                                                    |                                            |      | 2014 | AIP Handbook (Order 5100.38D) pavement LCCA table; also the 1992 A-94 benefit-cost base case reinstated by OMB M-25-23 (April 2025) for BCA, not for cost-effectiveness | <a href="https://www.whitehouse.gov/wp-content/uploads/2025/02/M-25-23-Rescissi">https://www.whitehouse.gov/wp-content/uploads/2025/02/M-25-23-Rescissi</a> |
| FAA AC 150/5320-6E Appendix 1 (superseded)                                    | 4% and 20 years; straight-line residual value                                      |                                            |      | 2009 | Dropped in 6F/6G; still widely cited                                                                                                                                    | <a href="https://www.faa.gov/documentlibrary/media/Advisory_Circular/150_5320_6">https://www.faa.gov/documentlibrary/media/Advisory_Circular/150_5320_6</a> |
| AATPT 06-06 airport LCCA guide (Auburn/NCAT)                                  | 4%, 20 years, NPW, remaining-life salvage; user cost includes lost airport revenue |                                            |      | 2011 | Basis for the "airport daily revenue" approach used here                                                                                                                | <a href="https://www.eng.auburn.edu/research/centers/hcat/files/aatpt/Report.FI">https://www.eng.auburn.edu/research/centers/hcat/files/aatpt/Report.FI</a> |
| Caltrans LCCA manual / RealCost                                               | 4% real; 20 / 35 / 55 years (CAPM / rehabilitation / new construction)             |                                            |      | 2013 | Agency and user costs reported as present value and EUAC; the Summary comparison block follows that layout                                                              | <a href="https://dot.ca.gov/-/media/dot-media/programs/maintenance/documents/of">https://dot.ca.gov/-/media/dot-media/programs/maintenance/documents/of</a> |
| Structural design life (FAARFIELD, AC 150/5320-6G)                            |                                                                                    | 20 years                                   |      | 2021 | design life, not service life                                                                                                                                           | <a href="https://www.faa.gov/documentlibrary/media/Advisory_Circular/150-5320-6">https://www.faa.gov/documentlibrary/media/Advisory_Circular/150-5320-6</a> |
| Salvage: HMA                                                                  |                                                                                    | 0.13 fraction of the last mill-and-overlay |      |      | live from Maintenance Policies D32 (policy: 2 of 16 years remaining)                                                                                                    | this workbook                                                                                                                                               |
| Salvage: PCC                                                                  |                                                                                    | 0.25 fraction of initial construction      |      |      | live from Maintenance Policies D46 (policy: 10 of 40 years remaining)                                                                                                   | this workbook                                                                                                                                               |
| Salvage method in FAA / AATPT guidance                                        | remaining life / expected life x cost of the last treatment (straight line)        |                                            |      | 2011 | the workbook policy follows this form; the fractions above are the policy decision flagged in the change notes                                                          | <a href="https://www.eng.auburn.edu/research/centers/hcat/files/aatpt/Report.FI">https://www.eng.auburn.edu/research/centers/hcat/files/aatpt/Report.FI</a> |

The Summary sensitivity chart runs 2% to 8% so both the current FAA rate (about 2%) and the legacy 7% are on the curve. G10 on the Summary states whether the lowest-cost alternative changes at either rate.

##### 6. Maintenance timing and service life (Maintenance Policies sheet against published experience)

| Item                                                  | Workbook policy (live)         | Published range                                                                       | Year |      | Reading                                                                                  | Source                                                                                                                                                      |
|-------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------|------|------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HMA surface treatment / markings                      | years 4, 8, 24                 | preventive-maintenance window PCI 60-80; inspect PCI every 3 years (AIP airports)     |      | 2014 | FAA ACs give the PCI window, not fixed year intervals; 4-year cycles are the TDOT policy | <a href="https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-7">https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-7</a> |
| HMA patching + crack sealing + treatment              | years 12, 16, 28               | crack sealing, patching, seal coats per AC 150/5380-6C                                |      | 2014 |                                                                                          | <a href="https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-6">https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-6</a> |
| HMA mill and overlay                                  | year 20                        | asphalt runways reach PCI 70 in about 12 to 15 years (FAA 2014 performance trends)    |      | 2014 | policy year 20 is at the optimistic end of the published range                           | <a href="https://www.acpa.org/wp-content/uploads/2025/03/ACPA-Achieving-the-Goa">https://www.acpa.org/wp-content/uploads/2025/03/ACPA-Achieving-the-Goa</a> |
| PCC joint resealing + patching                        | year 19                        | hot-pour sealant 3-8 yr; silicone 8-15 yr; compression seals 15-20 yr                 |      |      | policy year 19 assumes silicone-class sealant life                                       | <a href="https://www.tarmacview.com/glossary/joint-sealant/">https://www.tarmacview.com/glossary/joint-sealant/</a>                                         |
| PCC slab replacement (2%) + joints                    | year 27                        | PCC runways reach PCI 70 at about 40 years (FAA 2014)                                 |      | 2014 |                                                                                          | <a href="https://www.acpa.org/wp-content/uploads/2025/03/ACPA-Achieving-the-Goa">https://www.acpa.org/wp-content/uploads/2025/03/ACPA-Achieving-the-Goa</a> |
| Expected service life                                 | HMA 20 yr design; PCC 30-40 yr | FAA Extended Airport Pavement Life program targets 40 years                           |      | 2024 |                                                                                          | <a href="https://www.airporttech.tc.faa.gov/Portals/0/FactSheets/Extended%20Air">https://www.airporttech.tc.faa.gov/Portals/0/FactSheets/Extended%20Air</a> |
| Treatment fact sheets with life and cost by treatment |                                | ACRP Report 159 (GA airports) and ACRP Synthesis 22                                   |      | 2016 | use for project-specific overrides of the policy years                                   | <a href="https://nap.nationalacademies.org/catalog/23649/">https://nap.nationalacademies.org/catalog/23649/</a>                                             |
| Minimum layer thickness, AC 150/5320-6G               |                                | P-401 surface 4 in (>= 12,500 lb); P-209 base 6 in; P-501 6 in (5 in under 12,500 lb) |      | 2021 | check the pay-item quantities against these minimums                                     | <a href="https://www.faa.gov/documentlibrary/media/Advisory_Circular/150-5320-6">https://www.faa.gov/documentlibrary/media/Advisory_Circular/150-5320-6</a> |

Maintenance Policies drives every activity year and rate; this section only shows them next to the published ranges. Sources were read from search excerpts in September 2026 because the FAA and OMB sites could not be opened from the build environment; confirm the cited paragraphs before quoting them to FAA.

##### Airport daily revenue used for lost-revenue cost (the 17 airports with revenue data; live from RevenueData)

| Airport ID | Daily revenue (\$/day) | Lost revenue, 7-day closure (\$) | Lost revenue, 19-day closure (\$) | Columns holding text (counted as \$0) | Source                                      |
|------------|------------------------|----------------------------------|-----------------------------------|---------------------------------------|---------------------------------------------|
| MKL        | 8,505.94               | 59541.58                         | 161612.86                         | other                                 | RevenueData sheet in this workbook (hidden) |
| SRB        | 4,344.12               | 30408.84                         | 82538.28                          | other                                 | RevenueData sheet in this workbook (hidden) |
| MBT        | 8,841.38               | 61889.66                         | 167986.22                         | other                                 | RevenueData sheet in this workbook (hidden) |
| NQA        | 6,291.00               | 44037                            | 119529                            |                                       | RevenueData sheet in this workbook (hidden) |
| DKX        | 6,703.00               | 46921                            | 127357                            |                                       | RevenueData sheet in this workbook (hidden) |
| SYI        | 30,816.64              | 215716.48                        | 585516.16                         |                                       | RevenueData sheet in this workbook (hidden) |
| GCY        | 1,800.00               | 12600                            | 34200                             | tenant                                | RevenueData sheet in this workbook (hidden) |
| THA        | 939.26                 | 6574.82                          | 17845.94                          | tenant                                | RevenueData sheet in this workbook (hidden) |
| GKT        | 5,441.17               | 38088.19                         | 103382.23                         | tenant other                          | RevenueData sheet in this workbook (hidden) |
| JWN        | 89,240.00              | 624680                           | 1695560                           | tenant                                | RevenueData sheet in this workbook (hidden) |
| MOR        | 8,801.40               | 61609.8                          | 167226.6                          | tenant other                          | RevenueData sheet in this workbook (hidden) |
| MQY        | 8,841.38               | 61889.66                         | 167986.22                         | other                                 | RevenueData sheet in this workbook (hidden) |
| RZR        | 4,100.00               | 28700                            | 77900                             |                                       | RevenueData sheet in this workbook (hidden) |
| XNX        | 81.97                  | 573.79                           | 1557.43                           | tenant other                          | RevenueData sheet in this workbook (hidden) |
| M91        | 2,058.00               | 14406                            | 39102                             |                                       | RevenueData sheet in this workbook (hidden) |
| CKV        | 3,239.39               | 22675.73                         | 61548.41                          | tenant other                          | RevenueData sheet in this workbook (hidden) |
| M54        | 1,027.81               | 7194.67                          | 19528.39                          | 100LL Jet-A                           | RevenueData sheet in this workbook (hidden) |

Same SUMIF chain as General Information D39: text entries in RevenueData (" \$- ", " NA ", descriptions on XNX and M54) count as zero. 7 days is a typical surface-treatment closure and 19 days a mill-and-overlay closure for a 6,000 x 100 ft runway (see production rates).

| Airports in the General Information dropdown    |              |  |  |  |                                         |
|-------------------------------------------------|--------------|--|--|--|-----------------------------------------|
| Item                                            | Value        |  |  |  | Source                                  |
| Airports selectable in D9                       | 79.00        |  |  |  | General Information, hidden columns L:P |
| Airports with revenue data                      | 17.00        |  |  |  | RevenueData                             |
| Airports with a Middle / West / East region tag | 33 / 21 / 25 |  |  |  | General Information, hidden columns L:P |